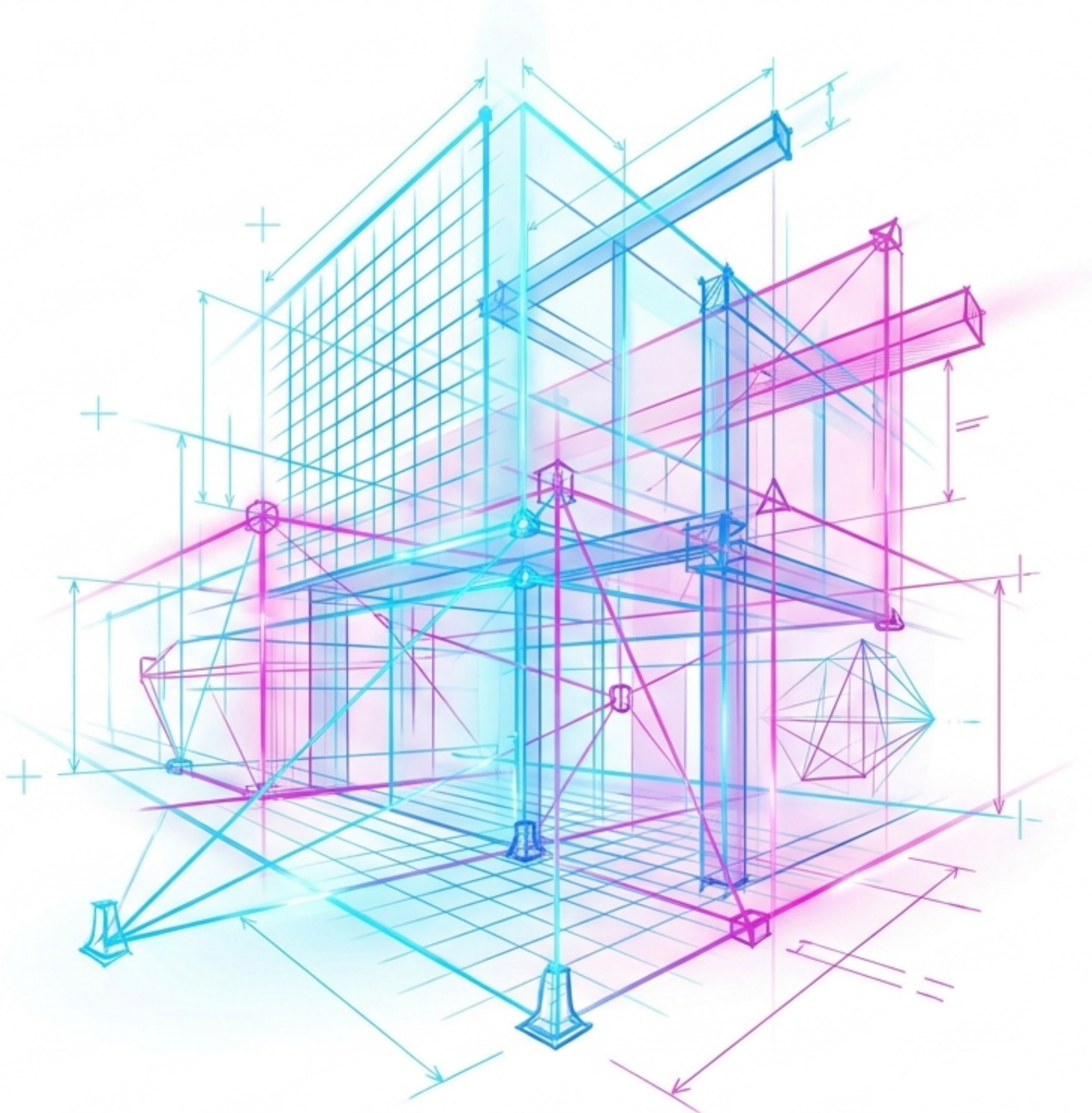
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Coordinate Geometry: The Blueprint of Space

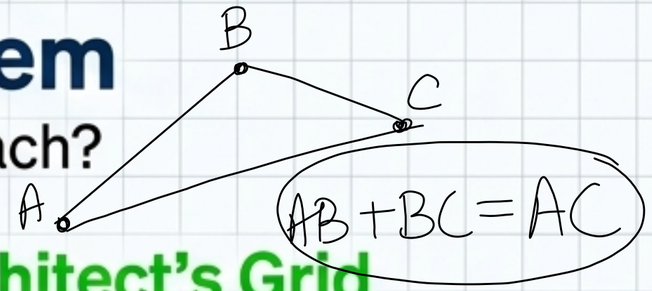
A Masterclass by Harshit Sir | Class X, Chapter 7



NotebookLM

The Real-World Spatial Problem

How do we measure what we cannot easily reach?



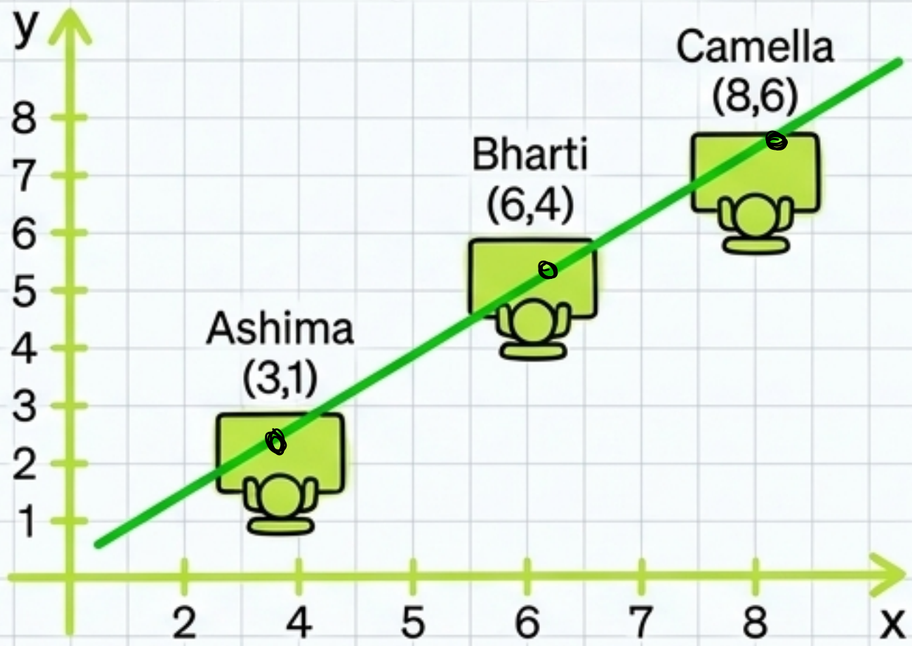
The Navigator's Dilemma

Town B is located 36 km east and 15 km north of Town A. How do we find the exact direct direct distance without a measuring tape?



The Architect's Grid

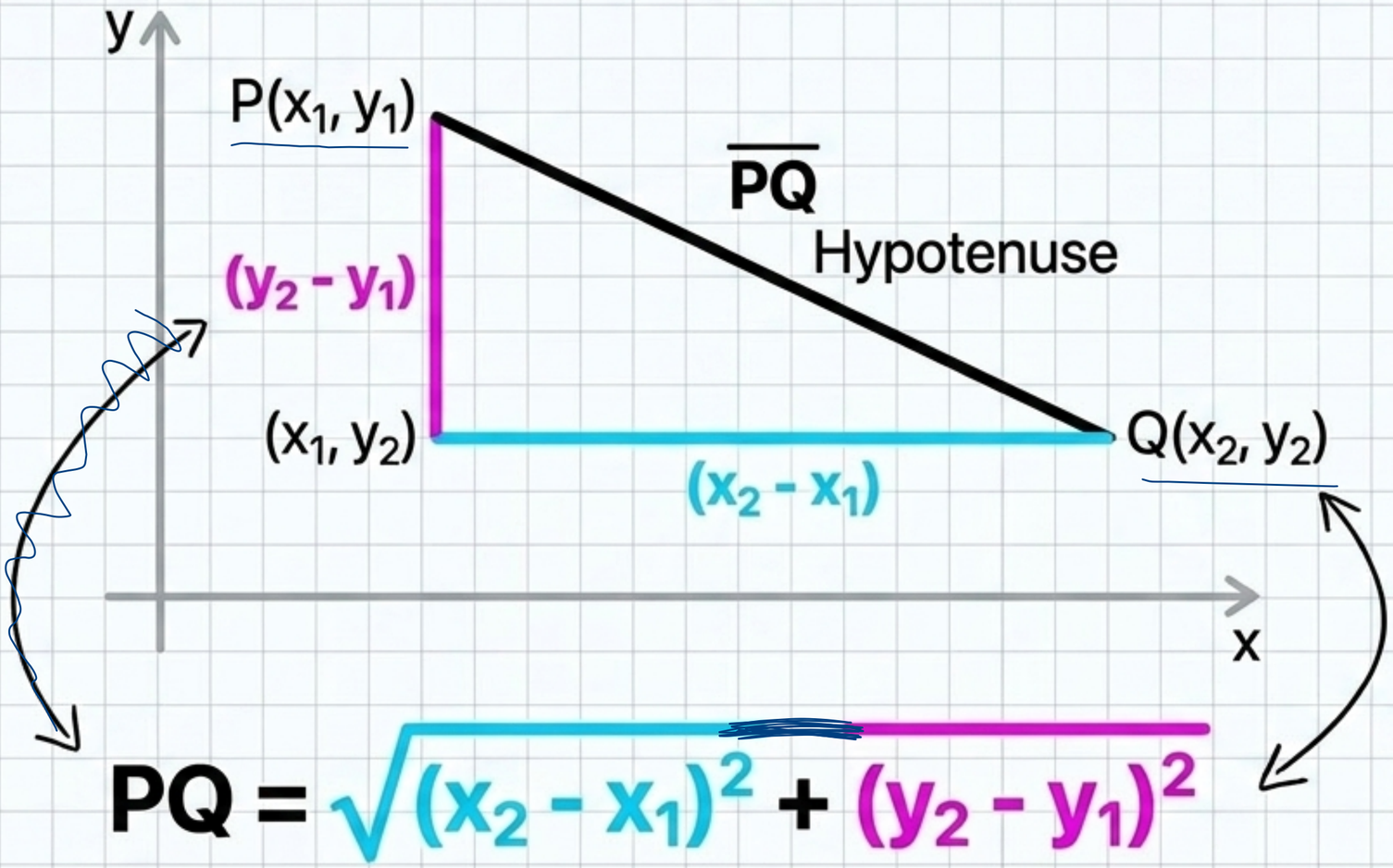
Ashima, Bharti, and Camella are seated at specific coordinates in a classroom. How can we mathematically prove they are sitting in a perfectly straight line?



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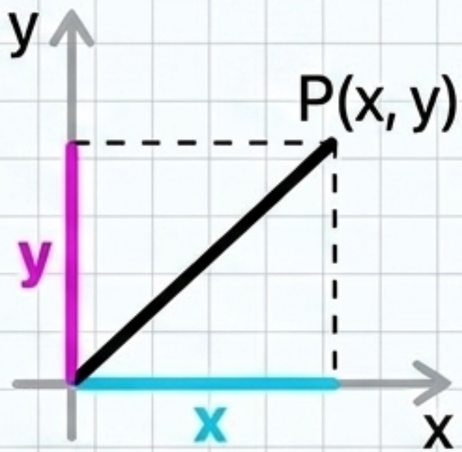
Tool #1: The Distance Formula

The distance between any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is just the hypotenuse of a right-angled triangle.

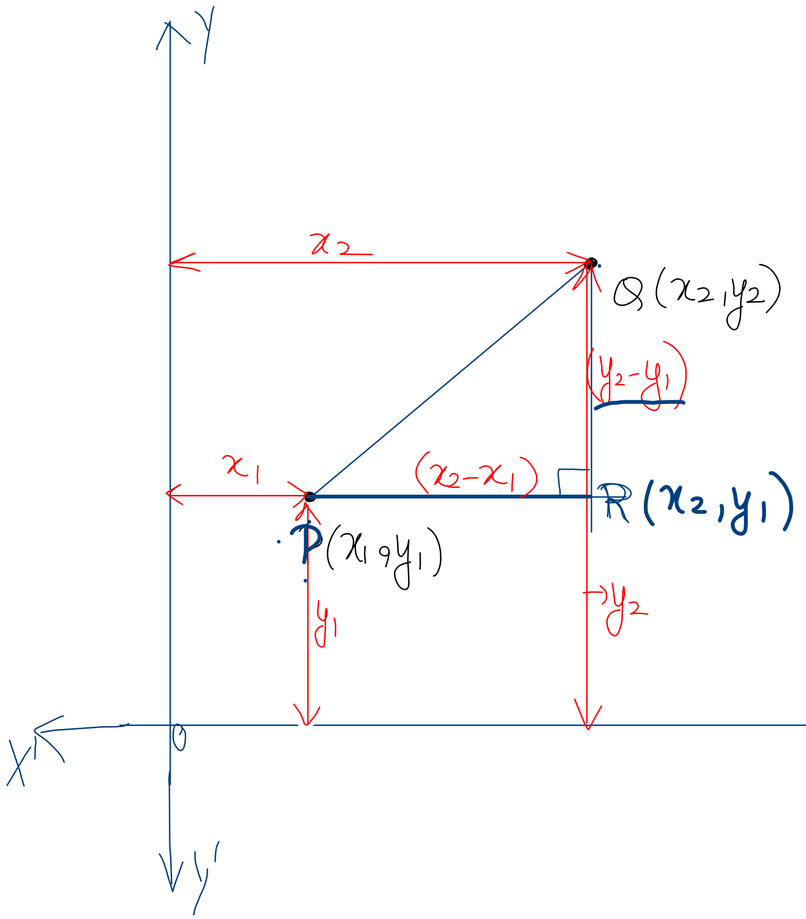


$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Special Case: The distance of a point P(x, y) from the origin O(0, 0) is simply $\sqrt{(x^2 + y^2)}$.



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To find the distance PQ.

Apply pythagoras theorem

$H^2 = P^2 + B^2$

$PQ^2 = (y_2-y_1)^2 + (x_2-x_1)^2$

$PQ^2 = (x_2-x_1)^2 + (y_2-y_1)^2$

Taking square root both side

$$PQ = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

Exercise 7.1

Q1. Find the distance between the following pair of points: (2, 3) and (4, 1).

Sol $(x_1, y_1) = (2, 3)$

$(x_2, y_2) = (4, 1)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(4-2)^2 + (1-3)^2}$$

$$= \sqrt{4 + 4}$$

$$= \sqrt{8} = 2\sqrt{2} \text{ units}$$

(a, b) and $(-a, -b)$

$(-5, 7)$ and $(-1, 3)$

x_1 y_1 x_2 y_2

$$\sqrt{[-1-(-5)]^2 + (3-7)^2}$$

$$= \sqrt{(-1+5)^2 + (-4)^2}$$

$$= \sqrt{16 + 16}$$

$$= \sqrt{32} = 4\sqrt{2} \text{ units}$$

Answer: $2\sqrt{2}$

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(a,b) and $(-a,-b)$

(x_1,y_1) (x_2,y_2)

$$\sqrt{(-a-a)^2+(-b-b)^2}$$

$$=\sqrt{(-2a)^2+(-2b)^2}$$

$$=\sqrt{4a^2+4b^2}$$

$$=\sqrt{4(a^2+b^2)}$$

$$=2\sqrt{a^2+b^2}$$

$(-a,-b)$ (a,b)

x_1, y_1 x_2 y_2

$$D=\sqrt{(a-(-a))^2+(b-(-b))^2}$$

$$=\sqrt{(2a)^2+(2b)^2}$$

$$=2\sqrt{a^2+b^2}$$

(0,0)

(3,4)

$(0,0)$ and (x,y)

$$D=\sqrt{x^2+y^2}$$

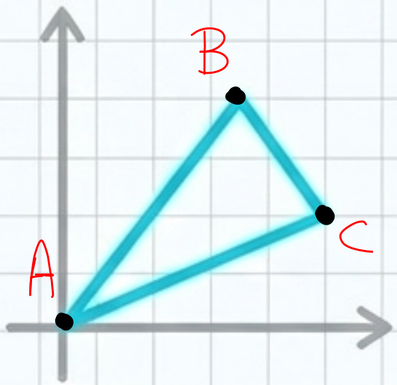
The distance between origin and (x,y) is

$$\sqrt{x^2+y^2}$$

Proving Geometric Shapes (Examples 1 & 2)

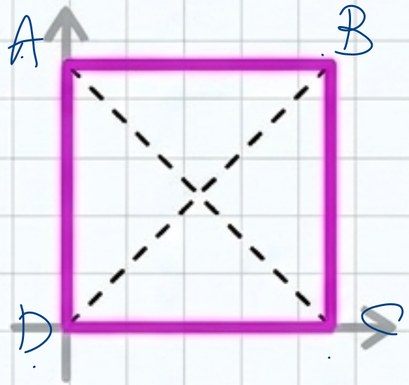
We use the distance formula to find side lengths and diagonals, then apply geometric rules to identify the figure.

To prove a Triangle (Example 1)



- Find lengths of all 3 sides.
- Ensure the sum of any two sides is greater than the third.

To prove a Square (Example 2)



- Find lengths of all 4 sides and both diagonals.
- Prove all 4 sides are equal AND both diagonals are equal.

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Qn 4. Check whether $(5,-2)$, $(6,4)$ and $(7,-2)$ are the vertices of an isoceles triangle.

$A = (5,-2)$

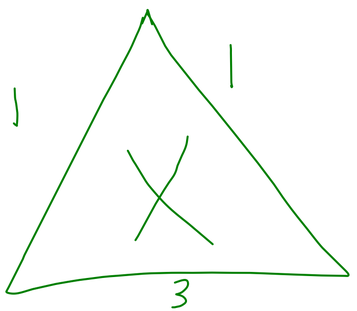
$B = (6,4)$]

$C = (7,-2)$

$AB =$

$BC =$

$AC =$



$$D = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Lines and Equidistance (Examples 3 & 4)

Example 3: Collinearity. Ashima (3, 1), Bharti (6, 4), and Camella (8, 6). Are they in a line?

Find AB, BC, and AC.

↓

If AB + BC = AC, they are collinear.

↓

Proof: $3\sqrt{2} + 2\sqrt{2} = 5\sqrt{2}$.
Yes, they are!

Example 4: Equidistance. Find a relation between x and y such that (x, y) is equidistant from (7, 1) and (3, 5).

Set AP = BP

↓

Square both sides and expand

↓

Simplify to get linear equation:
$x - y = 2$

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Intercepting the Axis (Example 5)

Problem: Find a point on the y-axis which is equidistant from A(6, 5) and B(-4, 3).

The Blueprint

1. Any point on the y-axis has an x-coordinate of 0. Let the point be P(0, y).

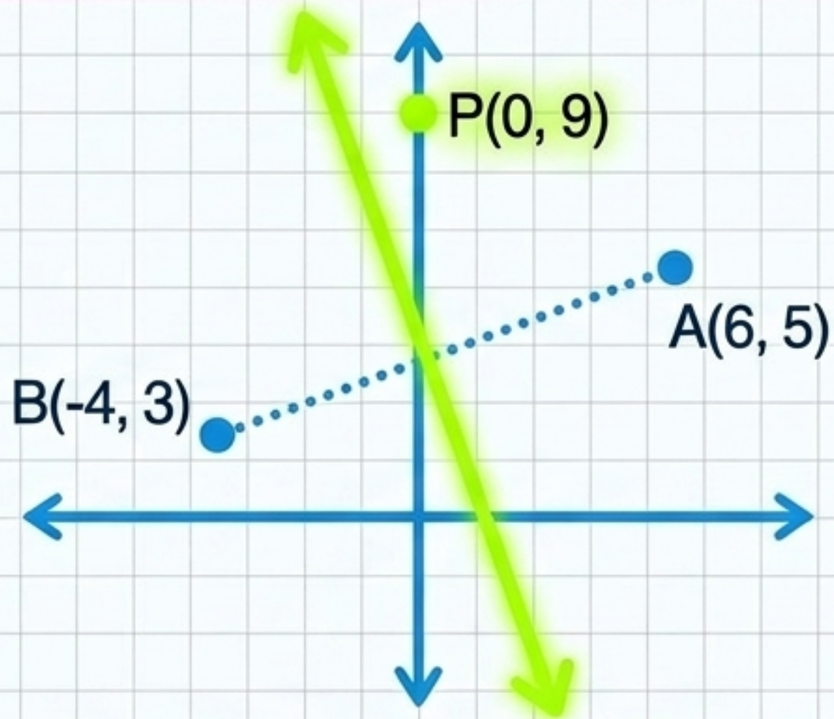
2. Set $PA^2 = PB^2$.

3. $(0 - 6)^2 + (y - 5)^2 = (0 - (-4))^2 + (y - 3)^2$

4. Simplify: $36 + y^2 - 10y + 25 = 16 + y^2 - 6y + 9$

5. Solve for y: $y = 9$. The point is (0, 9).

The Visual



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Exercise 7.1

Q7. Find the point on the x-axis which is equidistant from (2, -5) and (-2, 9).

(-7,0)

Let the point on x-axis is (a,0).

$$D=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$4+x^2+2(-2)(-x)$

$4+x^2+4x$

$4+x^2-2(-2)(x)$

$4+x^2+4x$

P(a,0)

(x_1,y_1)

(x_2,y_2)

(2,-5)

A

(-2,9)

B

$PA = PB$

$$\sqrt{(2-a)^2+(-5-0)^2}=\sqrt{(-2-a)^2+(9-0)^2}$$

$$\Rightarrow\sqrt{4+a^2-4a+25}=\sqrt{4+a^2+4a+81}$$

$$\Rightarrow a^2-4a+29=a^2+4a+85$$

$$-8a=56$$

$$a=-7$$

(-7,0)

Answer:(-7,0)

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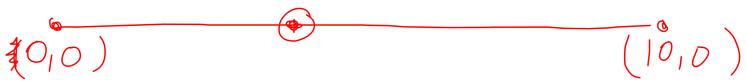
The Relay Tower Problem

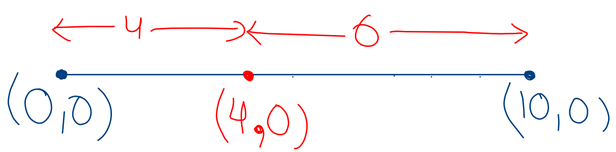
Suppose a telephone company wants to position a relay tower at point P between City A and City B. They need the distance of the tower from City B to be exactly twice its distance from City A. How do we find the exact coordinates for the tower without guessing?



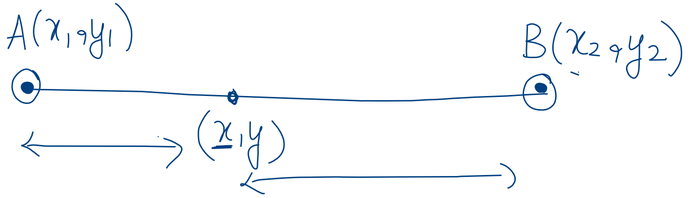
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Section formula





2:3



m:n

or

$m_1 : m_2$

Section formula

$$x = \frac{m(x_2)+n(x_1)}{m+n}$$

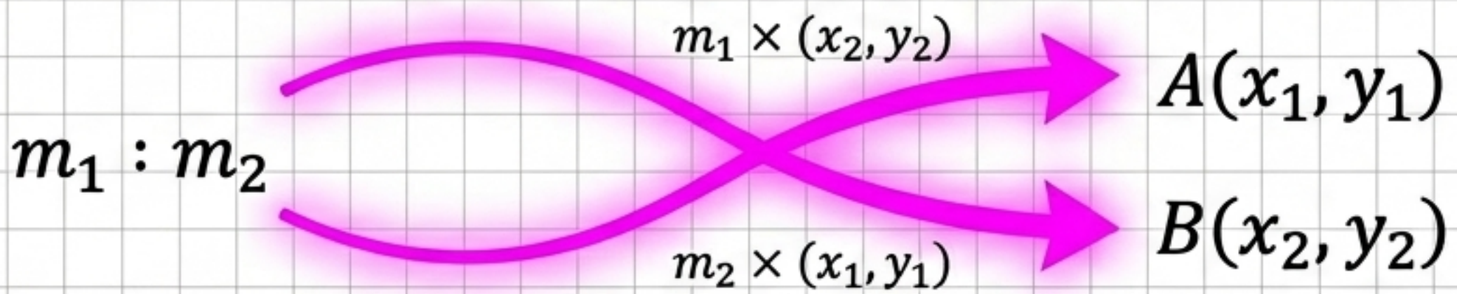
$$y = \frac{m(y_2)+n(y_1)}{m+n}$$

Tool #2: The Section Formula

To find the coordinates of point $P(x, y)$ that divides a line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ interally in the ratio $m_1 : m_2$:

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \qquad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

The Cross-Multiplication Trick



The Midpoint Shortcut: When a point divides a line exactly in half, the ratio is 1:1. The formula perfectly simplifies to: $x = \frac{x_1 + x_2}{2}$, $y = \frac{y_1 + y_2}{2}$

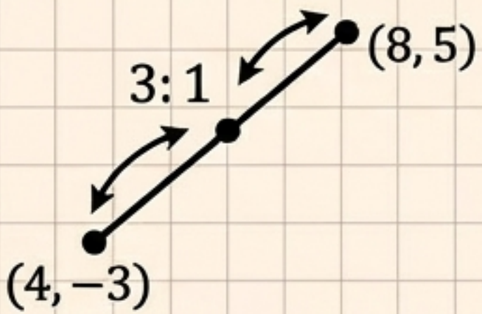
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Ratios and Coordinates (Examples 6 & 7)

Example 6 (Finding Coordinates)

Point divides $(4, -3)$ and $(8, 5)$ in ratio 3:1.

Apply formula directly to get $(7, 3)$.



Example 7 (Finding the Ratio - The k:1 Trick)

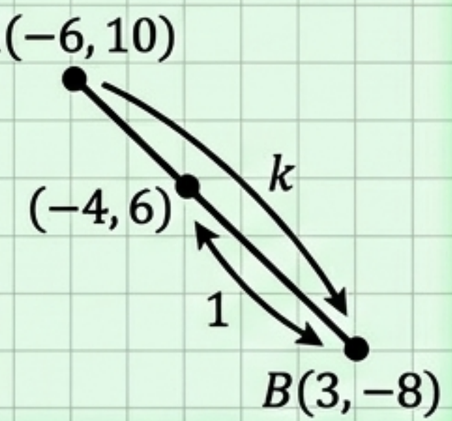
In what ratio does $(-4, 6)$ divide $A(-6, 10)$ and $B(3, -8)$?



Pro-tip: Instead of using $m_1:m_2$, assume the ratio is k:1. This leaves only one variable to solve!

Set $-4 = \frac{k(3) + 1(-6)}{k + 1}$

$\rightarrow k = \frac{2}{7}$. Ratio is 2:7.



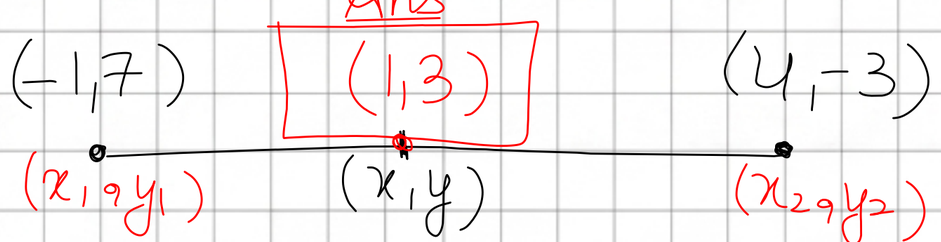
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Exercise 7.2

Q1. Find the coordinates of the point which divides the join of (−1, 7) and (4, −3) in the ratio 2 : 3.

Ans

$(1,3)$



$m : n = 2 : 3$

$$x = \frac{mx_2 + nx_1}{m+n} = \frac{2(4) + 3(-1)}{2+3} = \frac{5}{5} = 1$$

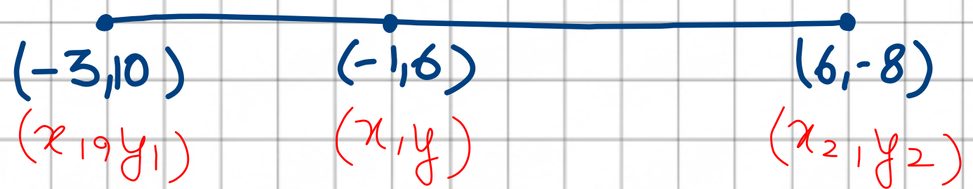
$$y = \frac{my_2 + ny_1}{m+n} = \frac{2(-3) + 3(7)}{2+3} = \frac{15}{5} = 3$$

Answer: (1, 3)

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Exercise 7.2

Q4. Find the ratio in which the line segment joining the points (-3, 10) and (6, -8) is divided by (-1, 6).



$$x = \frac{mx_2 + nx_1}{m+n}$$

$$-1 = \frac{m(6) + n(-3)}{m+n}$$

$$\Rightarrow \frac{-1}{1} = \frac{6m - 3n}{m+n}$$

$$6m - 3n = -m - n$$

$$6m + m = 3n - n$$

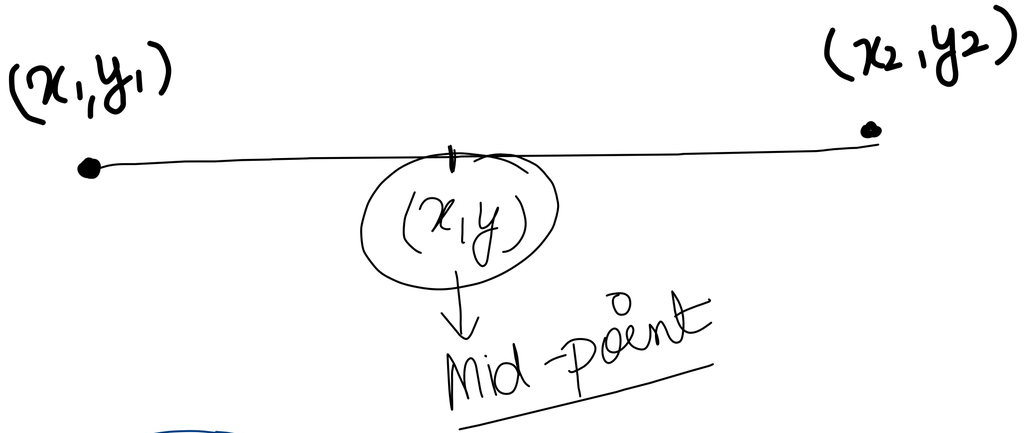
$$7m = 2n$$

$$\frac{m}{n} = \frac{2}{7}$$

$m:n = 2:7$

Answer: 2:7

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Mid-point formula $(x, y) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$m : n = 1 : 1$ $\quad m = n$

$$x = \frac{mx_2 + nx_1}{m+n} = \frac{x_2 + x_1}{2}$$

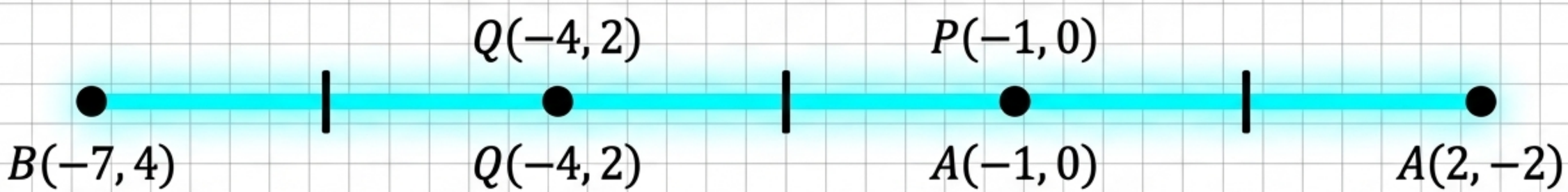
$$y = \frac{my_2 + ny_1}{m+n} = \frac{y_2 + y_1}{2}$$

Points of Trisection (Example 8)

Problem: Find the points of trisection of the line segment joining $A(2, -2)$ and $B(-7, 4)$.

The Blueprint:

- Trisection means dividing into three equal parts. This creates TWO points, P and Q.
  1. Point P divides $\overrightarrow{AB}$ in the ratio 1:2. Apply Section Formula → $P(-1, 0)$.
  2. Point Q divides $\overrightarrow{AB}$ in the ratio 2:1. (Or simply recognize Q is the midpoint of $\overrightarrow{PB}$). Apply formula → $Q(-4, 2)$.



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Advanced Blueprinting (Examples 9 & 10)

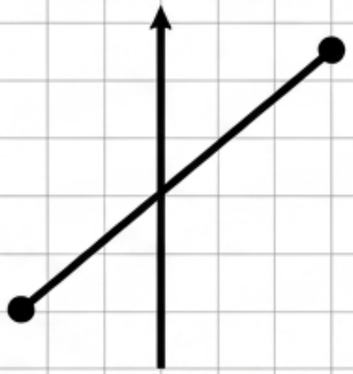
Example 9: The y-axis division.

Find the ratio the y-axis divides $(5, -6)$ and $(-1, -4)$. Use k:1.

Since it's the y-axis, the x-coordinate must be 0!

Set $\frac{k(-1) + 1(5)}{k + 1} = 0$

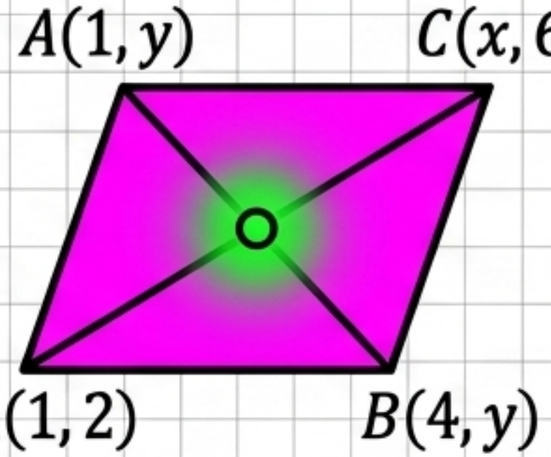
$\rightarrow k = 5$. Ratio is 5:1.



Example 10: The Parallelogram.

If $(1, 2), (4, y), (x, 6), (3, 5)$ are vertices of a parallelogram, find x and y.

Golden Rule:
Diagonals of a parallelogram bisect each other.



Midpoint of $\overrightarrow{AC}$ = Midpoint of $\overrightarrow{BD}$.
Solve for x and y easily!

NotebookLM

Independent Practice: Distance Formula

Q9. Find the point on the x-axis which is equidistant from (-2, 5) and (2, -3).

Answer: (-2, 0)

NotebookLM

Independent Practice: Distance Formula

Q17. Find the values of k for which the distance between the points A(k, -5) and B(2, 7) is 13 units.

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

x_1 y_1 x_2 y_2

$AB = 13$

$$\sqrt{(2-k)^2+(7+5)^2} = 13$$

$$\sqrt{4+k^2-4k+144} = 13$$

$$\Rightarrow \sqrt{k^2-4k+148} = 13$$

$$k^2-4k+148 = 169$$

$$k^2-4k = 21$$

$$\boxed{k^2-4k-21=0}$$

$$k^2-7k+3k-21=0$$

$$k(k-7)+3(k-7)=0$$

$$(k-7)(k+3)=0$$

$k-7=0$ OR $k+3=0$

$\boxed{k=7}$ or $\boxed{k=-3}$

Answer: k = 7 or k = -3

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Qn. If $Q(0,1)$ is equidistant from $P(5,-3)$ and $R(x,6)$ find the values of x. Also find the distance QR and PR.

Independent Practice: Section Formula

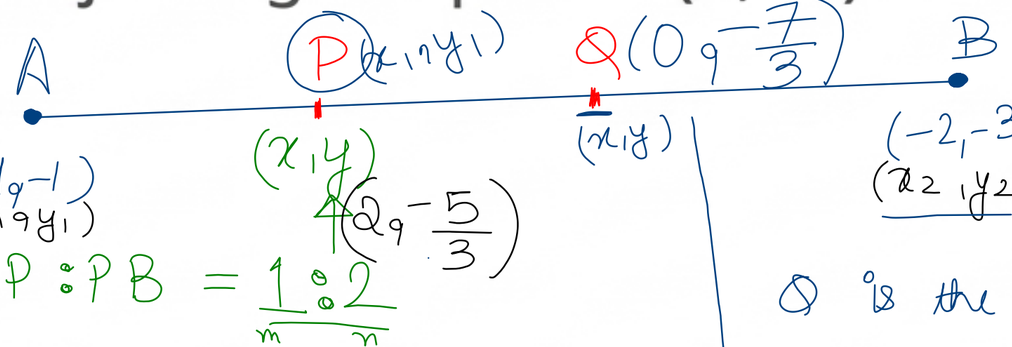
Q1. Find the coordinates of the point which divides the line segment joining the points A(4, -3) and B(9, 7) in the ratio 3 : 2.

Answer: (7, 3)

NotebookLM

Independent Practice: Section Formula

Q13. Find the coordinates of the points of trisection of the line segment joining the points (4, -1) and (-2, -3).



AP : PB = 1 : 2
m : n

$x = \frac{mx_2 + nx_1}{m+n} = \frac{1(-2)+2(4)}{1+2} = \frac{6}{3} = 2$

$y = \frac{my_2 + ny_1}{m+n} = \frac{-3+(-2)}{3} = -\frac{5}{3}$

Q is the mid-point of PB.

$x = \frac{2+(-2)}{2} = 0$

$y = \frac{\frac{-5}{3} - 3}{2} = \frac{\frac{-5-9}{3}}{2} = \frac{-14}{6} = -\frac{7}{3}$

Answer: (2, -5/3) and (0, -7/3)

NotebookLM

The Architect's Blueprint: Chapter 7 Synthesis

Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Use for: Lengths, Triangles, Quadrilaterals, Equidistance

Section Formula

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \qquad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

Use for: Internal division, Ratios using k:1

Origin Distance

$$d = \sqrt{x^2 + y^2}$$

Use for: Distance from (0,0)

Midpoint Formula

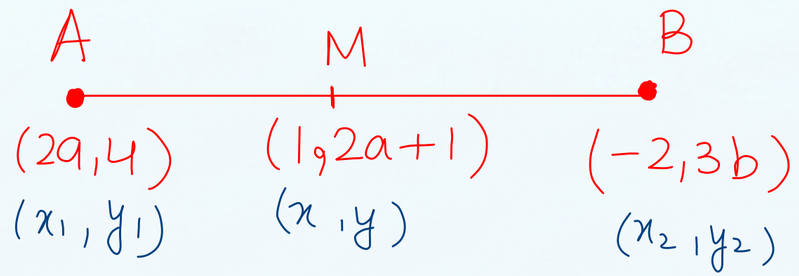
$$x = \frac{x_1 + x_2}{2} \qquad y = \frac{y_1 + y_2}{2}$$

Use for: Parallelogram diagonals, bisectors

NotebookLM

The Final Challenge

Q8. The midpoint of the line segment joining $A(2a, 4)$ and $B(-2, 3b)$ is $M(1, 2a + 1)$. Find the values of a and b.



$$x = \frac{x_1 + x_2}{2} \qquad y = \frac{y_1 + y_2}{2}$$

$$1 = \frac{2a - 2}{2} \qquad 2a + 1 = \frac{4 + 3b}{2}$$

$$2a - 2 = 2 \qquad 5 = \frac{4 + 3b}{2}$$

$$2a = 4 \qquad 10 = 4 + 3b$$

$$\boxed{a = 2} \qquad 6 = 3b \Rightarrow \boxed{b = 2}$$

Answer: a = 2, b = 2

NotebookLM

10. The point P(1, 2) divides the join of A(–2, 1) and B(7, 4) are in the ratio of

(a) 3 : 2 (b) 2 : 3 (c) 2 : 1 (d) 1 : 2

11. A point P divides the join of A(5, –2) and B(9, 6) are in the ratio 3 : 1. The coordinates of P are

(a) (4, 7) (b) (8, 4) (c) $(\frac{11}{2}, 5)$ (d) (12, 8)

12. What point on x – axis is equidistant from the points A(7, 6) and B(–3, 4)?

(a) (0, 4) (b) (–4, 0) (c) (3, 0) (d) (0, 3)

13. The distance of the point P(4, –3) from the origin is

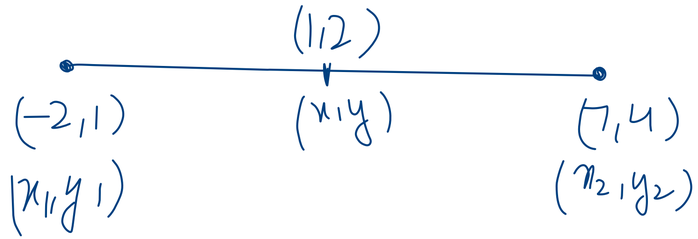
(a) 1 unit (b) 7 units (c) 5 units (d) 3 units

14. The distance between the points A(2, –3) and B(2, 2) is

(a) 2 units (b) 4 units (c) 5 units (d) 3 units

15. Find the area of the triangle whose vertices are A(1, 2), B(–2, 3) and C(–3, –4)

(a) 11sq. units (b) 22 sq. units (c) 7 sq. units (d) 6.5 sq. units



$$x = \frac{mx_2 + nx_1}{m+n}$$

$$1 = \frac{m(7) + n(-2)}{m+n}$$

$$1 = \frac{7m - 2n}{m+n}$$

$$7m - 2n = m + n$$

$$6m = 3n$$

$$\frac{m}{n} = \frac{3}{6} = \frac{1}{2}$$

$$\frac{27+5}{4} = \frac{32}{4} = 8$$

$$p8 + (-2) = \frac{16}{4} = 4$$